

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI- Semester of B.Sc. (Biochemistry) Examination April 2019

Course code & Name BC603: Enzyme Technology

Date: 29/04/2019 Day: Monday Time: 1: 30 PM To 1:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q I	Choose the correct answer for the following questions.		10
1.	To which class do Oxidases and dehydrogenases belong to?		
	(i) Transferases	(ii) Oxidoreductases	
	(iii) Ligases	(iv) None of the above	
2.	Which class of enzyme would catalyze a reaction where chemical formula does not change?		
	(i) Oxidoreductase	(ii) Transferase	
	(iii) Isomerase	(iv) Ligase	
3.	Speciality enzymes include the ones that are used in		
	(i) Pharmaceuticals	(ii) Diagnostics	
	(iii) Research	(iv) All of the above	
4.	A researcher plans to express an enzyme from human cell line that requires glycosylation for its optimum activity. The researcher clones cDNA encoding the enzyme in appropriate vector and expresses the enzyme in <i>E.coli</i> . Following purification he obtains very less activity in partially purified enzyme preparation. What could be the possible reason for it?		
	(i) Protein must have been degraded	(ii) <i>E.coli</i> does not produce glycosylation of proteins	
	(iii) Improper purification technique used for purification	(iv) He should have cloned mRNA rather than the cDNA	
5.	Which of the following methods can be used for directed evolution of enzyme		

	engineering?												
	(i) Error prone PCR	(ii) DNA -shuffling											
	(iii) Overlap extension PCR	(iv) Both i and ii											
6.	For purification of enzyme, enzymes are precipitated at their isoelectric point. Because enzymes are												
	(i) Most soluble at their pI	(ii) Least soluble at their pI											
	(iii) Remain unaffected at their pI	(iv) None of the above											
7.	Commonly used affinity tags that utilise a metal ligand for purification												
	(i) 6x- His	(ii) Maltose binding protein											
	(iii) GST tag	(iv) All of the above											
8.	Match the following												
	<table><tr><td>Industrial process</td><td>Enzyme used</td></tr><tr><td>1. Bioscouring</td><td>a. Proteases</td></tr><tr><td>2. Bating</td><td>b. Amylases</td></tr><tr><td>3. High –Fructose syrup</td><td>c. Hemi- cellulase</td></tr><tr><td>4. Wine making</td><td>d. Glucose isomerase</td></tr></table>			Industrial process	Enzyme used	1. Bioscouring	a. Proteases	2. Bating	b. Amylases	3. High –Fructose syrup	c. Hemi- cellulase	4. Wine making	d. Glucose isomerase
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	(i) 1- b; 2-a; 3-d; 4-c	(ii) 1-b; 2-d;3-a; 4-c											
	(iii) 1-c; 2-a; 3-d; 4-b	(iv) 1-b; 2-c; 3-a; 4-d											
9.	Thermolysin is used for the production of												
	(i) Aspartame	(ii) Aminopenicillinic acid											
	(iii) Lactose free milk	(iv) Racemic purification of amino acids											
10.	Engineering an enzyme with prior knowledge of structure and function of the enzyme is called												
	(i) Rational Design	(ii) Directed evolution											
	(iii) Either of the two	(iv) Neither of the two											

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VI- Semester of B.Sc. (Biochemistry) Examination April 2019

Course code & Name BC603: Enzyme Technology

Date: 29/04/2019 Day: Monday Time: 1: 50 PM To 3:30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I			10																					
Q - II Answer the following questions as directed																								
Q- II (A)	Describe the medical application of the following enzymes (Any two) a) Streptokinase b) Proteases such as Bromelain and Ficin c) Asparaginase		2																					
Q-II (B)	Describe briefly (Any two) a) Production of high fructose syrup b) Purification of cloned enzymes from <i>E.coli</i> c) Biosensors with an example		4																					
Q- II (C)	What are the different methods employed in enzyme engineering? Describe the methods that can be employed to generate engineered enzymes using directed evolution?		4																					
SECTION- II			15																					
Q- III (A)	From the information given below, which of the enzyme purification method has proved to be most effective? <table><tr><td>Purification steps</td><td>Protein (mg)</td><td>Enzyme activity (Units)</td></tr><tr><td>Crude extract</td><td>20000</td><td>4,000,000</td></tr><tr><td>Salt precipitation</td><td>5000</td><td>3,000,000</td></tr><tr><td>pH precipitation</td><td>4000</td><td>1,000,000</td></tr><tr><td>Ion exchange</td><td>200</td><td>800,000</td></tr><tr><td>Affinity chromatography</td><td>50</td><td>750,000</td></tr><tr><td>Size-exclusion</td><td>45</td><td>675,000</td></tr></table>	Purification steps	Protein (mg)	Enzyme activity (Units)	Crude extract	20000	4,000,000	Salt precipitation	5000	3,000,000	pH precipitation	4000	1,000,000	Ion exchange	200	800,000	Affinity chromatography	50	750,000	Size-exclusion	45	675,000		5
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Q-III (B)	a) What are immobilized enzymes? Describe any two methods of cell immobilization. OR b) Describe different enzyme reactors.		4																					
Q-III (C)	Describe the following processes and enzymes used in the process (any three) a) Biostoning b) Soaking c) Production of light beer d) Clarification of fruit juice.		6																					

